Venkata Lakshmi Parimala Pasupuleti

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github.com/venkatalakshmiparimala/AgentProbe | Remote - US

PROFESSIONAL SUMMARY

AI Engineer and systems builder with 9 years of production engineering and deep expertise in **RAG architectures, safety guardrails, and LLM observability**. Built AgentProbe -- implementing ChromaDB RAG pipelines, compliance-focused LLM-as-judge guardrails, prompt architecture for multi-agent reasoning, and a trace-level observability layer tracking retrieval accuracy and regression drift. Delivered HIPAA-compliant AI on structured healthcare records and production AI across 1.6M+ financial records with output validation. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 -- Aug 2025

Research Associate -- Virginia, USA

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets -- multi-source data collection and transformation; designed Tableau, ggplot2, and Quarto dashboards for faculty leadership improving comprehension by 40%.
- Applied Isolation Forest to automate data-quality checks across 3 research pipelines -- surfaced anomalies manual review missed, improving downstream model reliability.

Entrata
Jan 2017 -- Aug 2024

Software Engineer

- Applied Python, PyTorch, and OpenAI API to build AI pipelines across **1.6M financial records** with **output validation and quality gates** -- production AI on regulated data for enterprise clients at 100K+ MAU.
- Built 20+ full-stack microservices (Python, PostgreSQL, RabbitMQ, FastAPI) across 120+ Agile sprints; led Docker containerization cutting execution time 50%; improved query performance 30% across 200+ SQL queries.
- Achieved 100% PHI compliance on AWS S3 migration with Libsodium-encrypted pipelines -- zero security incidents; built affordable housing compliance modules (Section 8/HCV, LIHTC, HUD) - income eligibility verification, AMI threshold calculations, and HUD regulatory reporting across 100K+ unit portfolios; eliminated 80% manual workload via Python/PL-SQL automation across enterprise client environments.

PROJECTS

AgentProbe: LLM Eval + Agent Observability Toolkit
Open Source

Python · LangChain · LangGraph · FastAPI · ChromaDB · Anthropic · OpenAI · React · Docker · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

- Built **safety guardrails + observability layer**: compliance metric in LLM-as-judge (5-metric eval engine) blocks non-compliant agent outputs before production; React dashboard tracks retrieval accuracy, latency, and regression drift across concurrent agent test suites.
- Architected ChromaDB RAG pipeline with per-run retrieval quality metrics; 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, state handoffs, and multi-provider model routing -- deployed as 15-endpoint FastAPI backend.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker -- github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% by building an NLP platform translating natural language queries (EN/ES) to FHIR APIs -- **HIPAA-compliant RBAC** + AI governance across structured healthcare records; demonstrates safe AI deployment for end users in a regulated domain.

PropLens: Lease Intelligence & Compliance AI
In Progress

Python · LangGraph · ChromaDB · FastAPI · LLMs · React

- Building agent-based RAG pipeline over lease PDFs -- LLM-as-judge scoring, anomaly detection, and compliance gap detection across 1,000+ tenant records; extends observability and safety patterns from AgentProbe to a domain-specific compliance use case.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — Graduated May 2026
GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- Jawaharlal Nehru Technological University, Anantapur — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LLM Observability, RAG Pipelines, Prompt Engineering, Safety Guardrails, Agent Observability, LangChain, LangGraph, ChromaDB, Embeddings, LLM Evaluation, LLM-as-Judge, Eval Frameworks, OpenAI API, Anthropic API, Hugging Face, FastAPI

ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, BERT, spaCy, NumPy, Pandas, Tableau, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, PostgreSQL, MySQL